

BACKLOG SUBJECT EXAMINATIONS – AUGUST / SEPTEMBER 2023

B.E. - CSE (Cyber Security) /			
Program	: CSE (Artificial Intelligence and Machine Learning)	Semester	: III
Course Name	: Data Structures	Max. Marks	: 100
Course Code	: CY33 / CI33	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- A program has specified the size of array while declaring as intstuent ID [10], in one case the program has wasted the space by using only 5 values in the above array and in other case the program tries to store 15 values and fails to store more values than the size of array. Discuss how to overcome the above problem with the syntax for each. CO1 (08)
 - Explain the process of implementing sparse matrix using array representation. CO1 (08)
 - What is Abstract Data Types? Write the ADT of operations on Array. CO1 (04)
- In a small company there are five salesmen. Each salesman is supposed to sell three products. Write a program using a 2D array to print (i) the total sales by each salesman and (ii) total sales of each item. CO1 (08)
 - Write a program to input two m * n matrices and then calculate the sum of their corresponding elements and store it in a third m * n matrix. CO1 (08)
 - What is the significance of realloc()? Write the program to illustrate the operation of realloc(). CO1 (04)

UNIT - II

- Write a program to perform Push and Pop operations on a stack using array representation. CO2 (08)
 - Illustrate different steps involved in converting the following infix expression into postfix expression using stacks:

$$A - (B / C + (D \% E * F) / G) * H$$
CO2 (08)
 - Convert the following non-tailed recursive program to find the sum of n natural numbers where n >= 1 to tailed recursive program CO2 (04)
- ```
// Find sum of n natural number
int sum(int n)
{
 if(n == 1) {
 return n;
 }
 return n + sum(n - 1);
}
int main()
{
 int n = 4;
 cout << sum(n);
 return 0;
}
```

4. a) Illustrate how to evaluate the following prefix expression using stacks: CO2 (08)  
-\*+2(Two)2(Two)3(Three)10(Ten)  
b) Illustrate different steps involved in converting the following infix CO2 (08)  
expression into prefix expression using stacks A\*B-C/D+E.  
c) Write the program to find the factorial of a given number using tailed CO2 (04)  
recursive function.

## UNIT – III

5. a) Using diagrams explain the working of an ordinary queue for the CO3 (08)  
following sequence of operations  
-insert 5 elements continuously  
-delete 2 elements continuously  
-insert 4 elements continuously  
-display the status of the queue.  
b) What is the limitation of a linear queue? How to overcome this using CO3 (08)  
circular queue? Consider a circular queue of size 5 with 4 elements, 15,  
4, 23 and 6, where F=3 and R=1. Insert 50 and then delete two  
elements and finally insert 100. With necessary diagrams, clearly show  
the initial queue status with F and R values and queue status after each  
operation.  
c) What is a priority queue? Write the function module to insert elements to CO3 (04)  
the priority queue.
6. a) A Circular Queue whose size is 5 has 3 elements 245, 360, 115, where CO3 (06)  
F = 2 and R = 4 insert 450, 570 and delete one element from the queue.  
Show the sequence of steps with necessary diagrams indicating the  
values of F and R for each operation.  
b) Develop Qinsert() and Qdelete() functions for linear queue to CO3 (06)  
demonstrate insertion and deletion operations respectively.  
c) Implement a double ended queue with no restrictions (Insertion and CO3 (08)  
Deletion from both ends of the queue).

## UNIT- IV

7. a) Write a c program to create faculty database using singly linked CO4 (10)  
list whose information field stores following data about a faculty: name,  
emp\_id, salary, department. Display the details of the faculty who is  
drawing highest salary in cse department.  
b) Write the C Program to Delete the last node from a circular doubly linked CO4 (10)  
list.
8. a) Write a C program to create doubly linked list: CO4 (10)  
i. By inserting a node from rear end.  
ii. Deleting a node from front  
iii. Display the contents of the list from last node to first node.  
b) Write a C function to CO4 (10)  
i. To perform insertion of given node at specified position.  
ii. To perform deletion of a given node at specified position in doubly  
linked list.

## UNIT - V

9. a) Discuss the features of binary tree. Illustrate the traversal of binary tree CO5 (07)  
with an example.  
b) Describe the list representation of a tree with suitable example. CO5 (07)  
c) Explain different types of tree with an example for each. CO5 (06)

10. a) Convert the given general tree as shown in Fig. 10(a) to binary tree. CO5 (06)

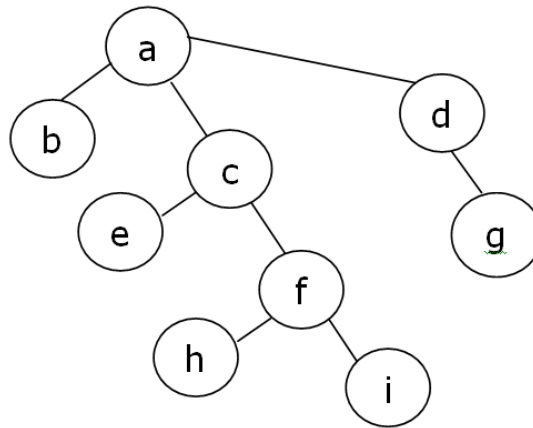


Fig.10(a) – Tree

- b) Explain the level order traversal of a binary tree with an example. CO5 (06)  
c) Construct the binary search tree stepwise from its inorder (CBDEAFHGI) and postorder (CEDBHIGFA) traversals. CO5 (08)

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